

Important questions viva ai

Instructions to AI

1. What are the major areas of AI?

Major areas include Machine Learning, Natural Language Processing (NLP), Robotics, Computer Vision, and Expert Systems.

2. Applications of AI

AI is used in fields like healthcare, finance, autonomous vehicles, virtual assistants, gaming, and fraud detection.

3. Agent in AI

An agent is an entity that perceives its environment and takes actions to achieve specific goals.

Search Techniques

4. What are search techniques in AI, and why are they important?

Search techniques help find solutions to problems by exploring possible actions. They are crucial for decision-making in AI.

5. Explain the difference between uninformed (blind) and informed search techniques.

Uninformed searches do not use problem-specific knowledge (e.g., BFS, DFS), while informed searches use heuristics to guide the search (e.g., A*).

6. What is the A search algorithm, and how does it work?*

A* is an informed search algorithm that uses a heuristic function to find the most optimal path by combining the cost to reach a node and the estimated cost to the goal.

7. What is the difference between depth-first search (DFS) and breadth-first search (BFS)?

DFS explores as deep as possible in each path before backtracking, while BFS explores all nodes at the current level before moving to the next level.

8. What is a heuristic function, and how does it affect search algorithms in AI?

A heuristic function estimates the cost to reach the goal from a given node, helping to guide search algorithms toward the most promising paths.

Knowledge Representation

9. What is knowledge representation in AI, and why is it crucial?

Knowledge representation is how information is structured and stored in AI, enabling reasoning and decision-making.

10. Explain the difference between declarative and procedural knowledge.

Declarative knowledge states facts or information, while procedural knowledge defines the steps to perform tasks or actions.

11. What are semantic networks, and how are they used in knowledge representation?

Semantic networks are graph structures that represent concepts as nodes and their relationships as edges, helping in understanding knowledge.

12. What is an ontology in AI, and how does it support knowledge representation?

An ontology defines a set of concepts and their relationships in a specific domain, allowing shared understanding and reasoning.

Uncertainty

13. How is uncertainty handled in AI systems?

Uncertainty is handled using probabilistic methods, fuzzy logic, or Bayesian networks to make informed decisions.

14. What is the difference between probabilistic reasoning and deterministic reasoning?

Probabilistic reasoning deals with uncertain outcomes using probabilities, while deterministic reasoning assumes certain outcomes.

Fuzzy Sets and Fuzzy Logic

15. What is a fuzzy set, and how does it differ from a classical set?

Fuzzy sets allow partial membership of elements, unlike classical sets, where an element is either fully in or out.

16. Explain the concept of fuzzy logic and its applications in AI.

Fuzzy logic deals with reasoning that is approximate rather than exact, used in control systems, decision-making, and pattern recognition.

17. How are membership functions used in fuzzy logic to represent uncertainty?

Membership functions define the degree to which an element belongs to a fuzzy set, indicating partial truth values.

Natural Language Processing (NLP)

18. What is Natural Language Processing (NLP), and what are its main components?

NLP is the technology for understanding human language, with components like tokenization, parsing, and semantic analysis.

19. Explain the difference between syntax and semantics in NLP.

Syntax refers to the structure of sentences, while semantics deals with the meaning of words and sentences.

Neural Networks

20. What is a neural network, and how does it mimic the human brain?

A neural network is a computational model inspired by the human brain, consisting of interconnected nodes (neurons) that process information.

21. What are the different layers in a neural network, and what are their roles?

Layers include the input layer (receives data), hidden layers (process data), and output layer (provides the result).

Expert Systems

22. What is an expert system, and how is it different from traditional AI systems?

An expert system emulates human decision-making using a knowledge base and inference rules, unlike traditional AI that relies on algorithms.

23. What are the main components of an expert system?

The main components are the knowledge base, inference engine, and user interface.

24. Explain the role of the inference engine in an expert system.

The inference engine processes information from the knowledge base to draw conclusions and make decisions.